EE 604 Image Processing

Lec 02: Spatial Domain - I

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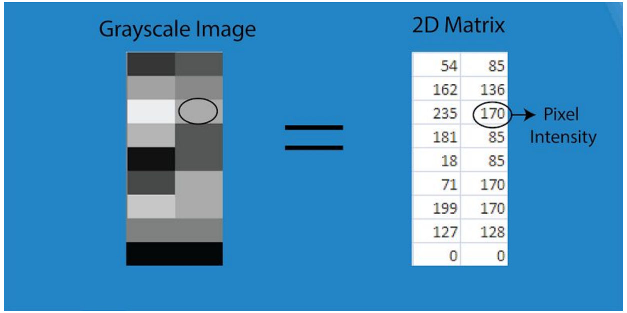
- Spatial Domain
- Sampling & Quantization
- Image as a matrix
- Image Processing in Spatial Domain
- Intensity Transformations

Spatial Domain

- **In image processing, the spatial domain refers to the image space itself, where operation and analysis are performed directly on the raw pixels of an image.**
- **Sampling & Quantization digitize a continuous real-world scene into the discrete 2D matrix of pixels that defines spatial domain processing.**



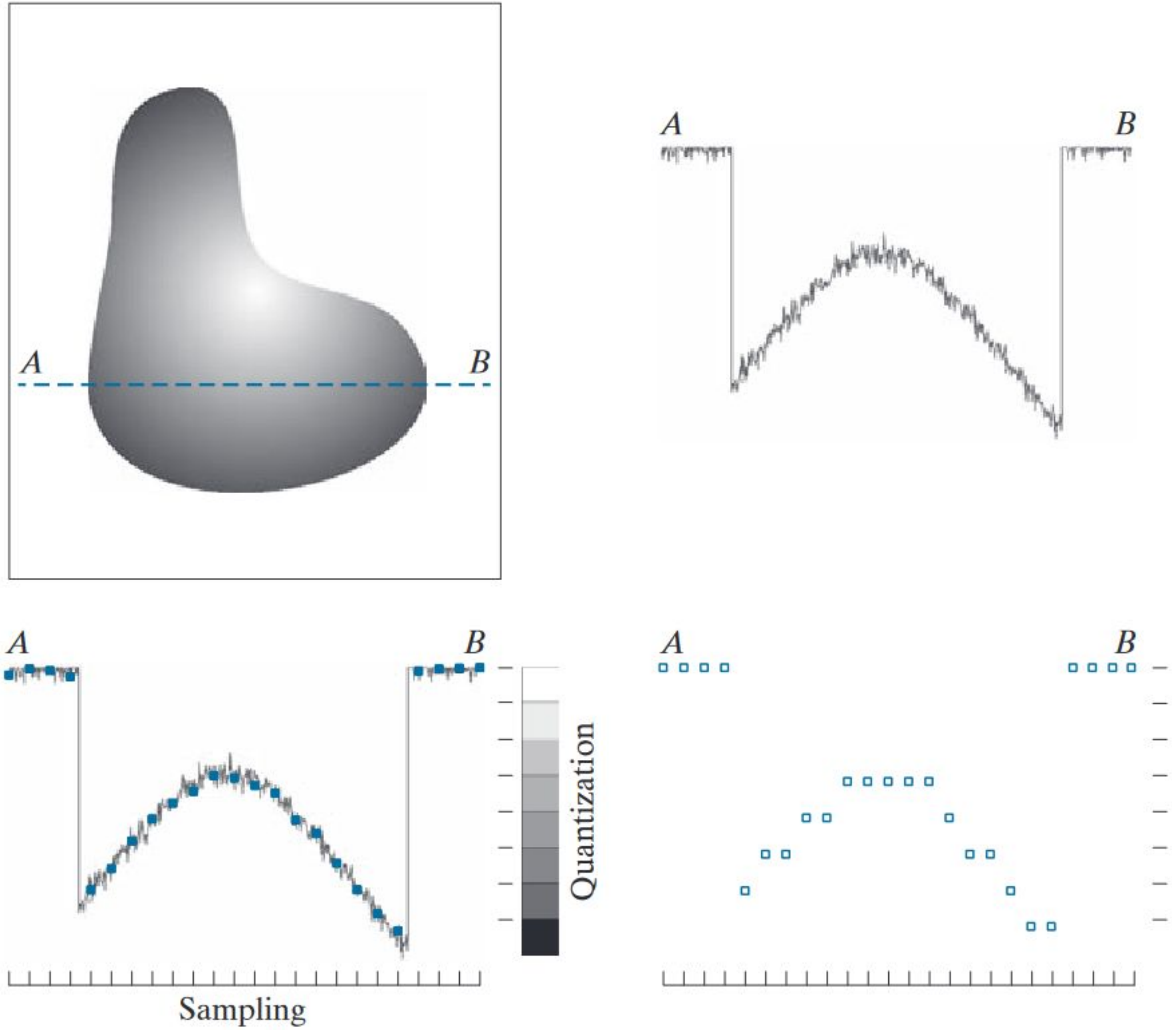
Spatial domain: Image mapped as a 2D matrix of pixel intensities

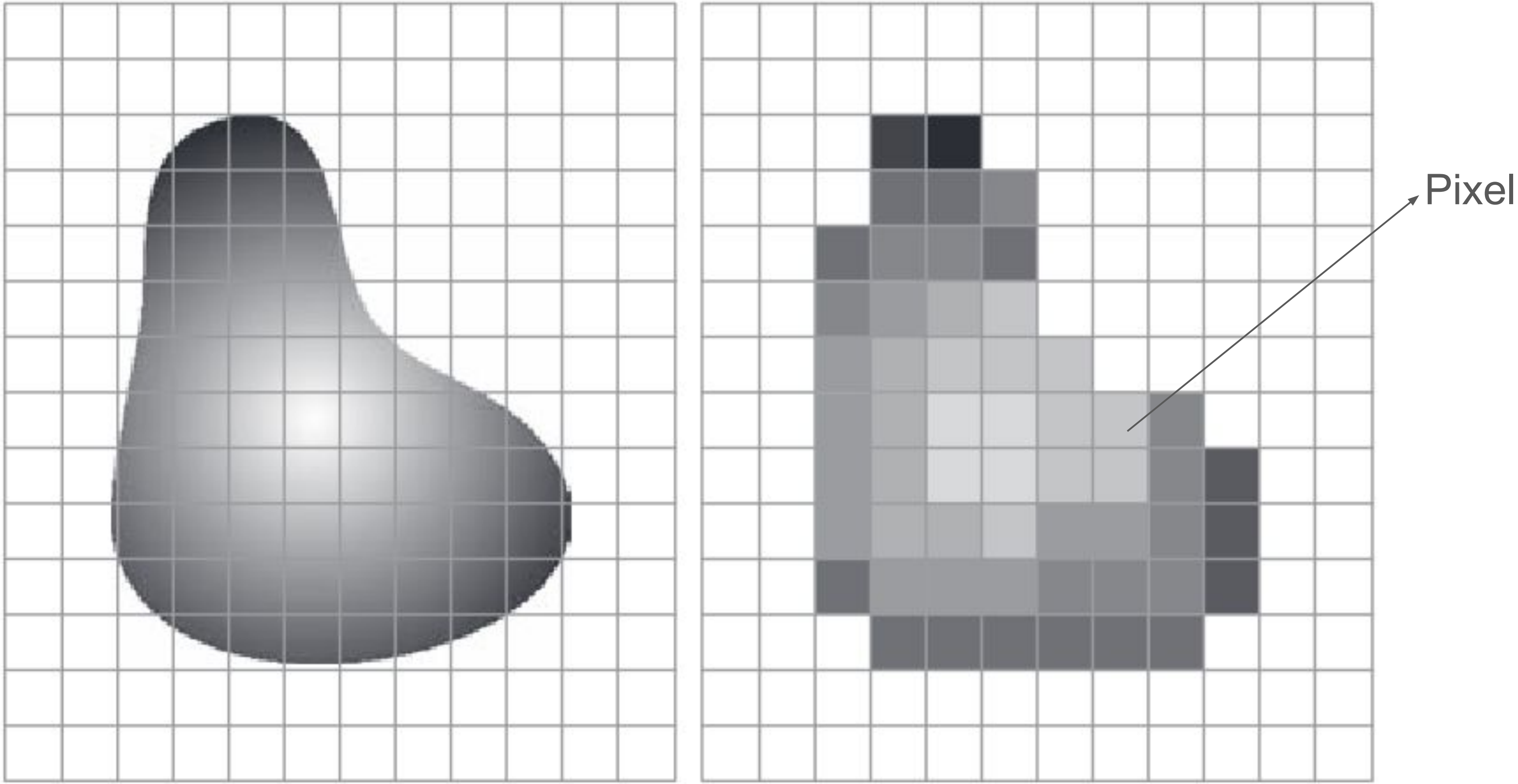
Sampling & Quantization

To have a digital image, we have to sample the function in both coordinates and also in amplitude.

Digitizing the coordinate values is called **sampling**.

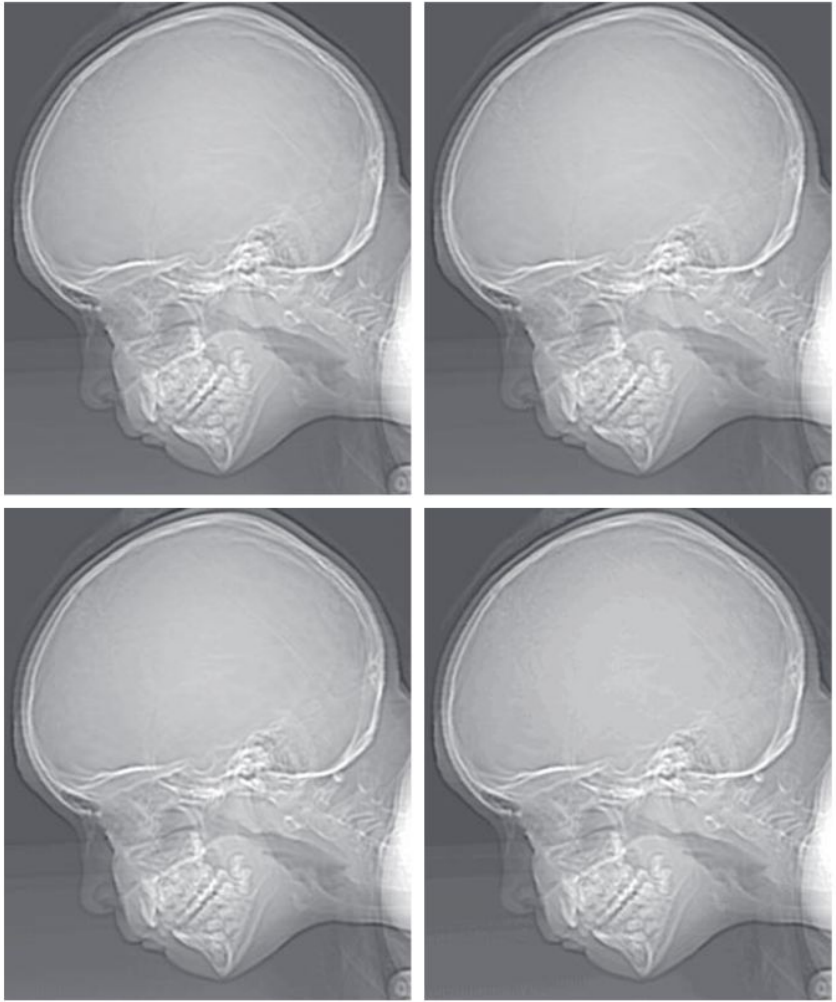
Digitizing the amplitude values is called **quantization**.




Pixel

a b
c d

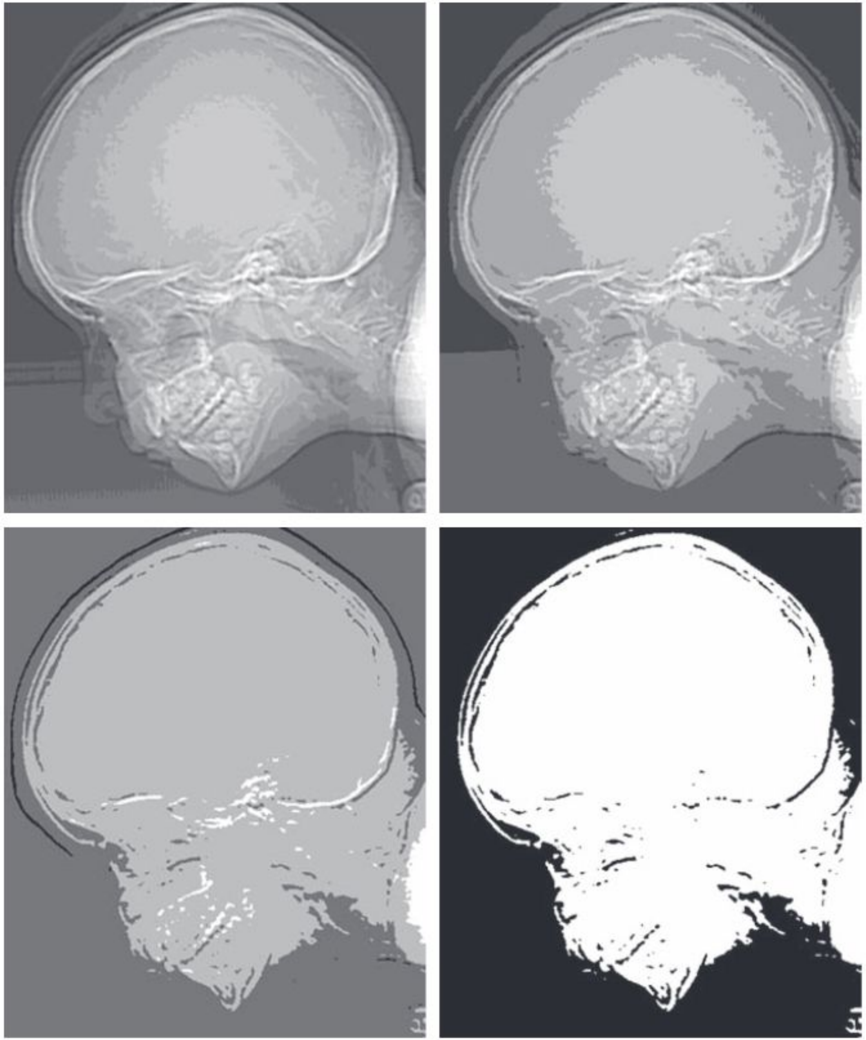
Effects of reducing spatial resolution. The images shown are at:
(a) 930 dpi,
(b) 300 dpi,
(c) 150 dpi, and
(d) 72 dpi.



a b
c d

(a) 774×640, 256-level image. (b)-(d) Image displayed in 128, 64, and 32 intensity levels, while keeping the spatial resolution constant. (Original image courtesy of the Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)

e f
g h



(*Continued*)
(e)-(h) Image displayed in 16, 8, 4, and 2 intensity levels.

Image as a matrix

In general, the value of a digital image at any coordinates (x,y) is denoted by f(x,y), where x and y are integers.

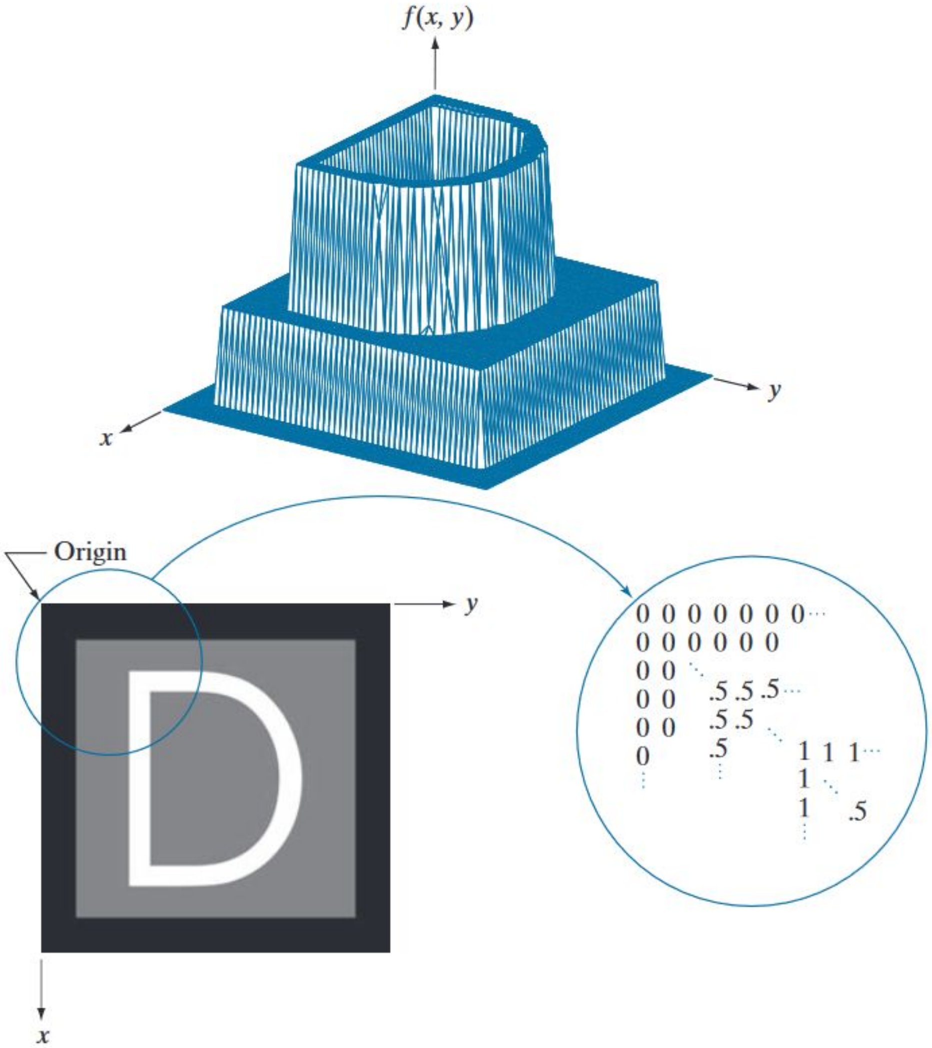
$$\begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

Image

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}$$

Pixel value at (1,N-1)

Clearly, $a_{ij} = f(i,j)$



a
b c

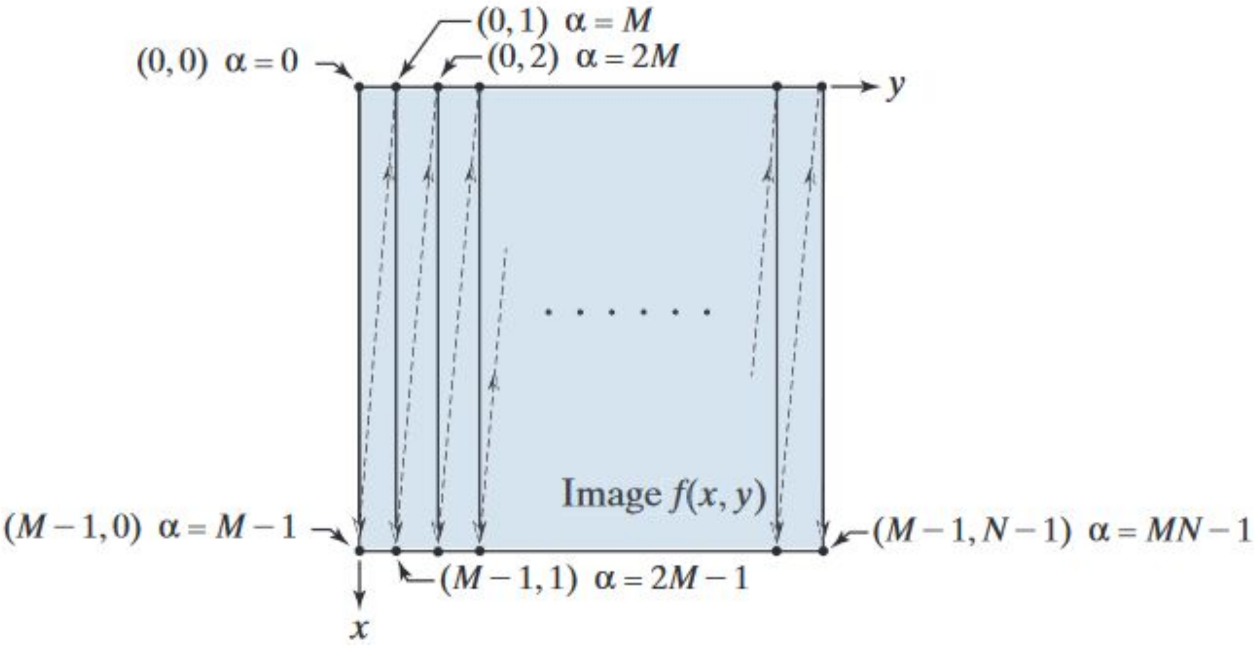
(a) Image plotted as a surface. (b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)

Image digitization requires that decisions be made regarding the values for M, N, and for the number, L, of discrete intensity levels. There are no restrictions placed on M and N, other than they have to be positive integers. However, digital storage and quantizing hardware considerations usually lead to the number of intensity levels, L, being an integer power of two; that is

$$L = 2^k$$

where k is an integer. We assume that the discrete levels are equally spaced and that they are integers in the range $[0, L-1]$.

Illustration of column scanning for generating linear indices. Shown are several 2-D coordinates (in parentheses) and their corresponding linear indices.



$$\alpha = My + x$$

$$x = \alpha \bmod M$$

$$y = (\alpha - x)/M$$

Image Processing in Spatial Domain

- Image processing methods in spatial domain are based on direct manipulation of pixels in an image.
- Principal categories of spatial processing:
  - Intensity Transformations
  - Spatial Filtering
- Intensity transformations operate on single pixels of an image for tasks such as contrast manipulation and image thresholding.
- Spatial filtering performs operations on the neighborhood of every pixel in an image. Examples of spatial filtering include image smoothing and sharpening.

Image enhancement: a good example of image processing in spatial domain

- Manipulate an image such that the result is more suitable than the original for viewing purposes or for a specific application.

- A method that is quite useful for enhancing X-ray images may not be the best approach for enhancing infrared images.

- The viewer is the ultimate judge of how well a particular method works, which is difficult to quantify.

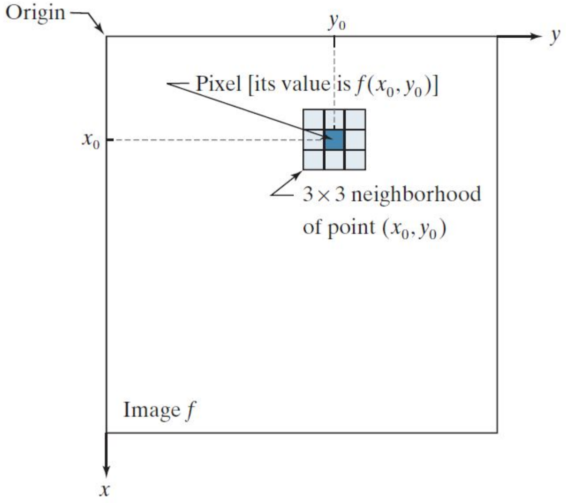
- However, in machine perception, enhancement is easier to quantify. In automated character recognition system, the most appropriate enhancement method would be the one with the best recognition rate.

The spatial domain processes we discuss in this lecture are based on the expression

$$g(x, y) = T[f(x, y)]$$

where $f(x, y)$ is an input image, $g(x, y)$ is the output image, and T is an operator on f defined over a neighborhood of point (x, y).

A 3×3 neighborhood about a point (x_0, y_0) in an image. The neighborhood is moved from pixel to pixel in the image to generate an output image.



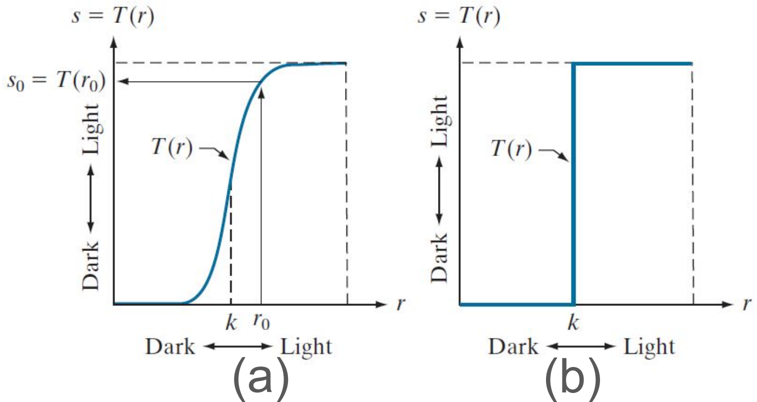
The smallest possible neighborhood is of size 1×1. In this case, g depends only on the value of f at a single point (x, y)

T becomes an *intensity transformation function* of the form

$$s = T(r)$$

where, for simplicity in notation, we use s and r to denote, respectively, the intensity of g and f at any point (x, y).

Intensity transformation functions.
(a) Contrast stretching function.
(b) Thresholding function.



Intensity Transformations

IMAGE NEGATIVES

LOG TRANSFORMATIONS

POWER-LAW (GAMMA) TRANSFORMATIONS

COMPLEX FUNCTIONS

Image Negatives

$$s = L - 1 - r$$

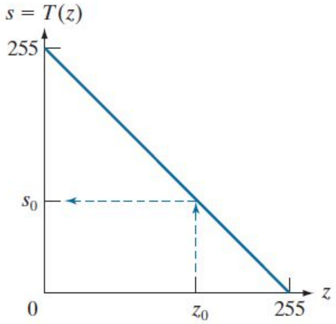
L=number of intensity levels

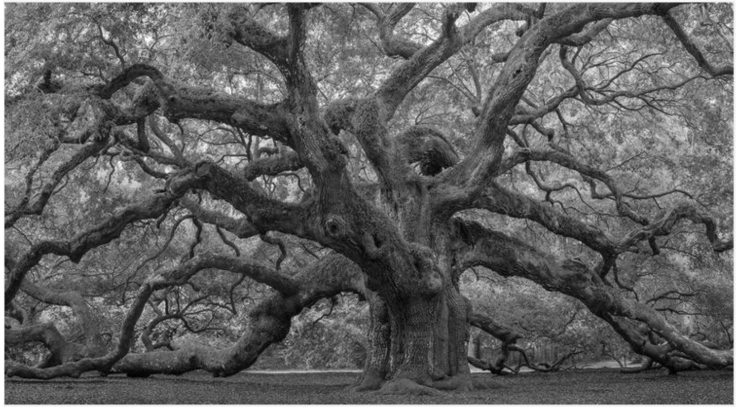
r=pixel intensity level of input image

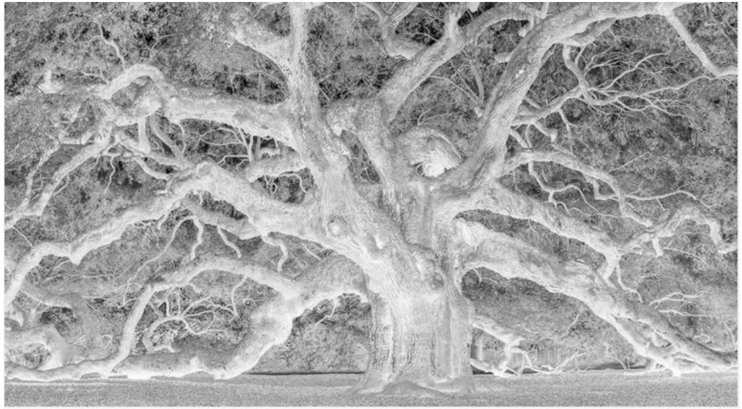
s=output intensity level of output image

Darkest pixels become brightest; brightest pixels become darkest.

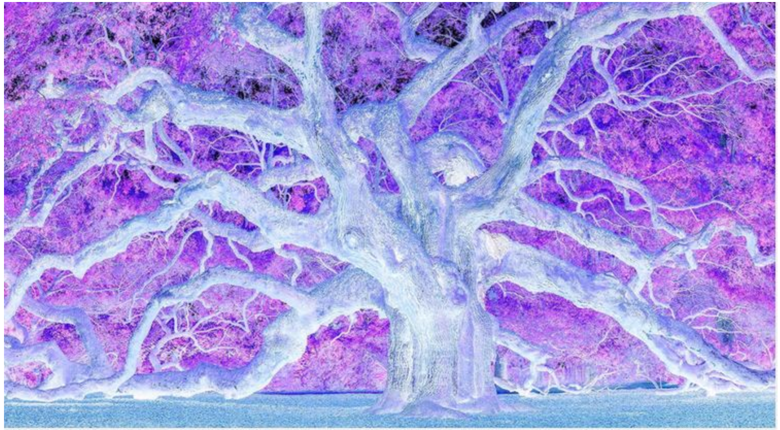
Intensity transformation function used to obtain the digital equivalent of photographic negative of an 8-bit image..





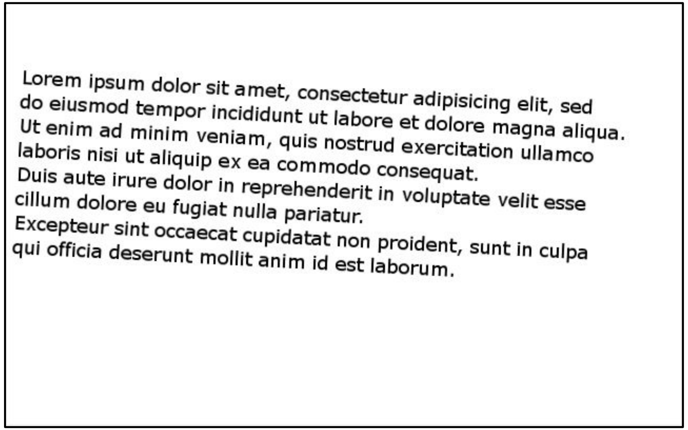


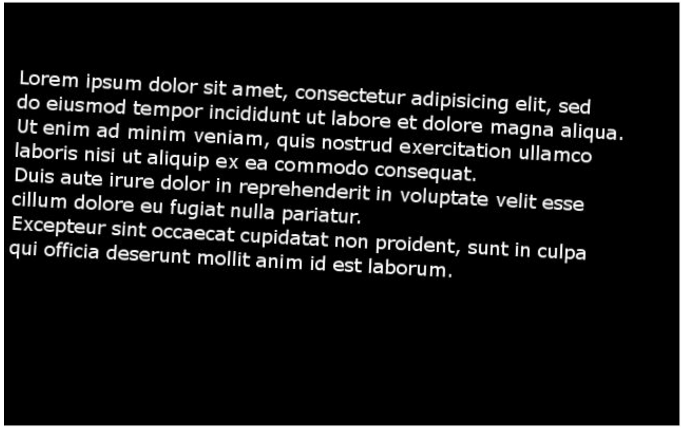
We can do it for color images also by applying it on individual channels



Applications

Text image processing

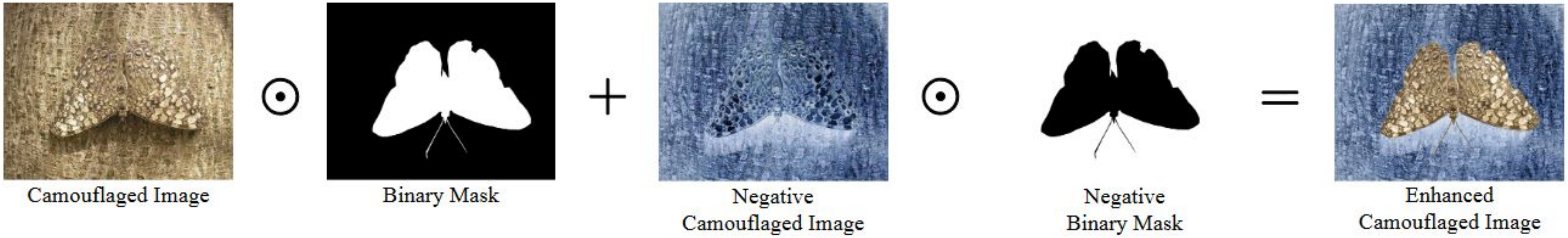




Text becomes the foreground

Another application

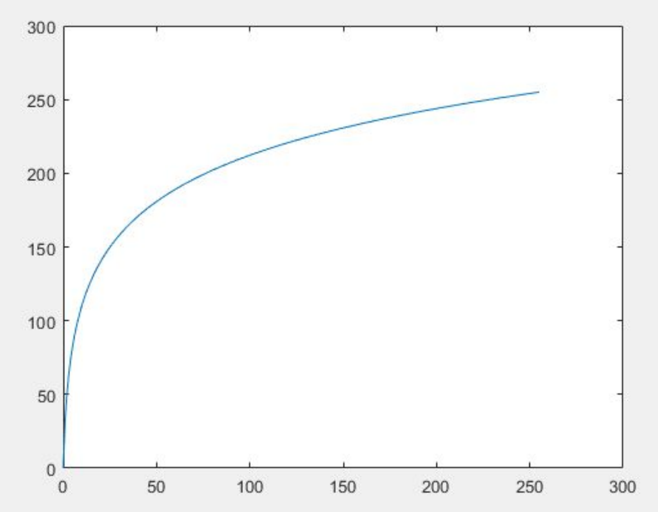
Enhancing camouflaged objects



A. Gupta, K. R. Jerripothula and T. Tillo, "CIRCOD: Co-Saliency Inspired Referring Camouflaged Object Discovery," *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson, AZ, USA, 2025, pp. 8313-8323, doi: 10.1109/WACV61041.2025.00806.

Log Transforms

$$s = c \log(1 + r)$$



c= a constant

r=pixel intensity level of input image

s=output intensity level of output image

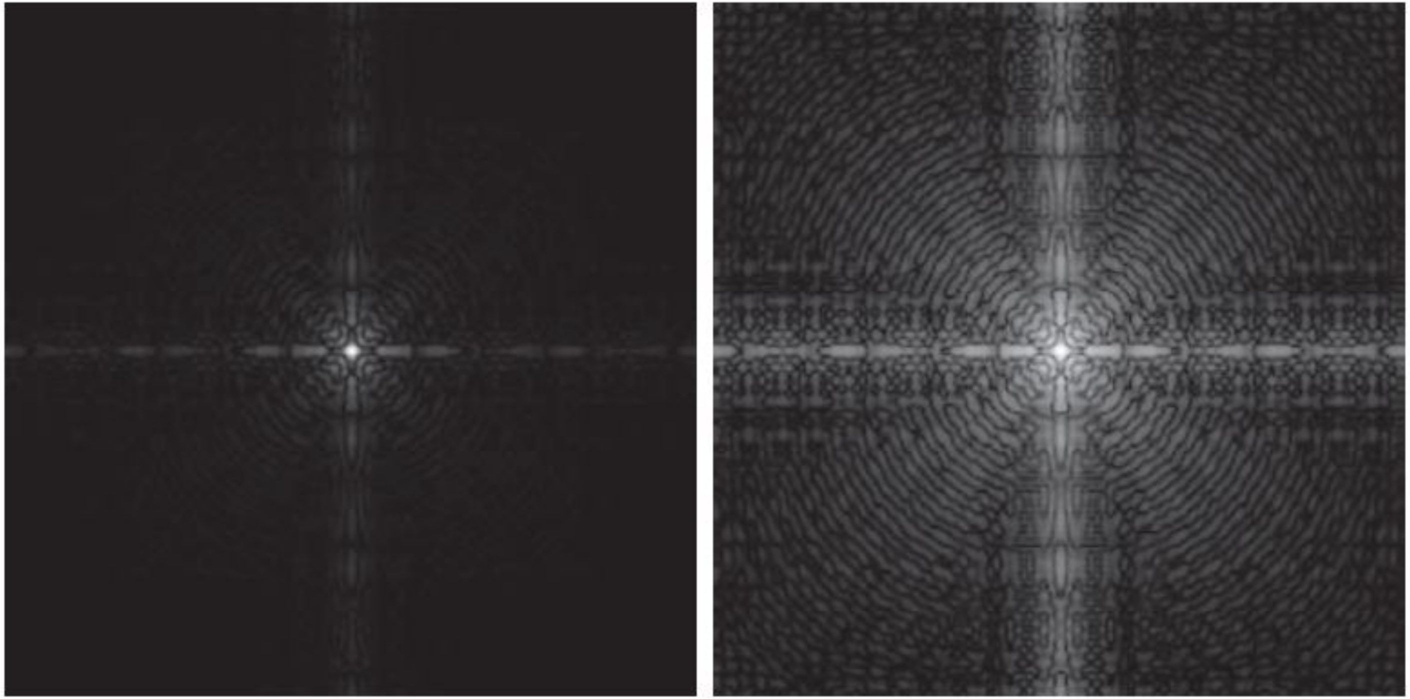
It maps a narrow range of low intensity values in the

input image into a wider range of output levels.

Conversely, higher values of input levels are mapped to a narrower range in the output.

Here,
c=255/log(256)

(a) Fourier spectrum displayed as a grayscale image. (b) Result of applying the log transformation with $c = 1$. Both images are scaled to the range $[0, 255]$.



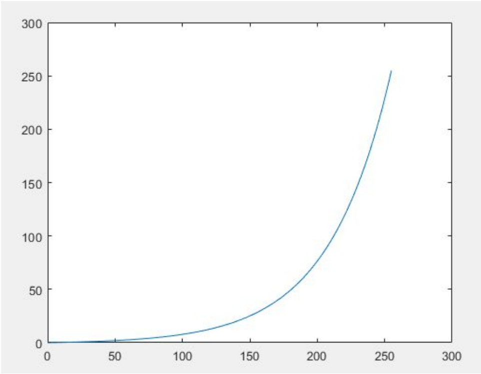
(a) (b)

There is also something called inverse log transformation:

s=exp(r/c)-1

Basically, it's the inverse function of the log transformation function.

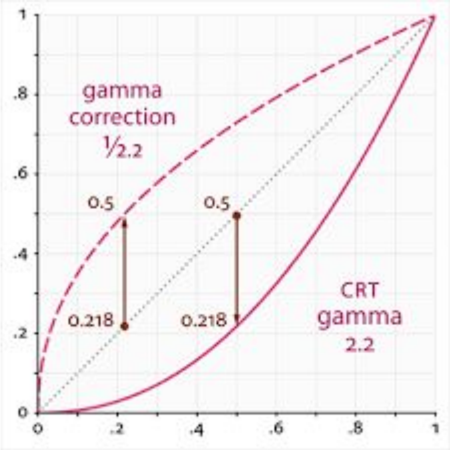
It gives opposite results: compresses lower values; expands higher values



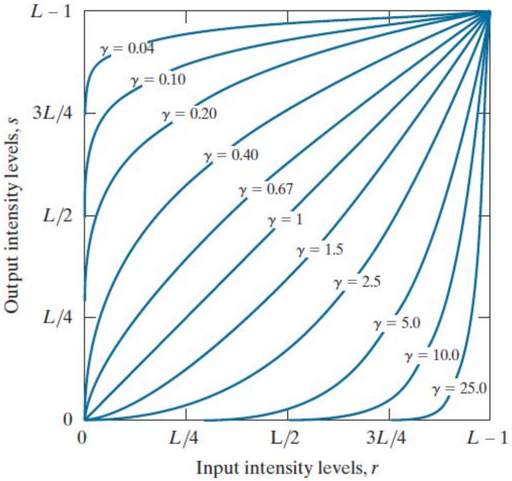
Here,
c=255/log(256)

POWER-LAW (GAMMA) Transformation

$$s = cr^{\gamma}$$

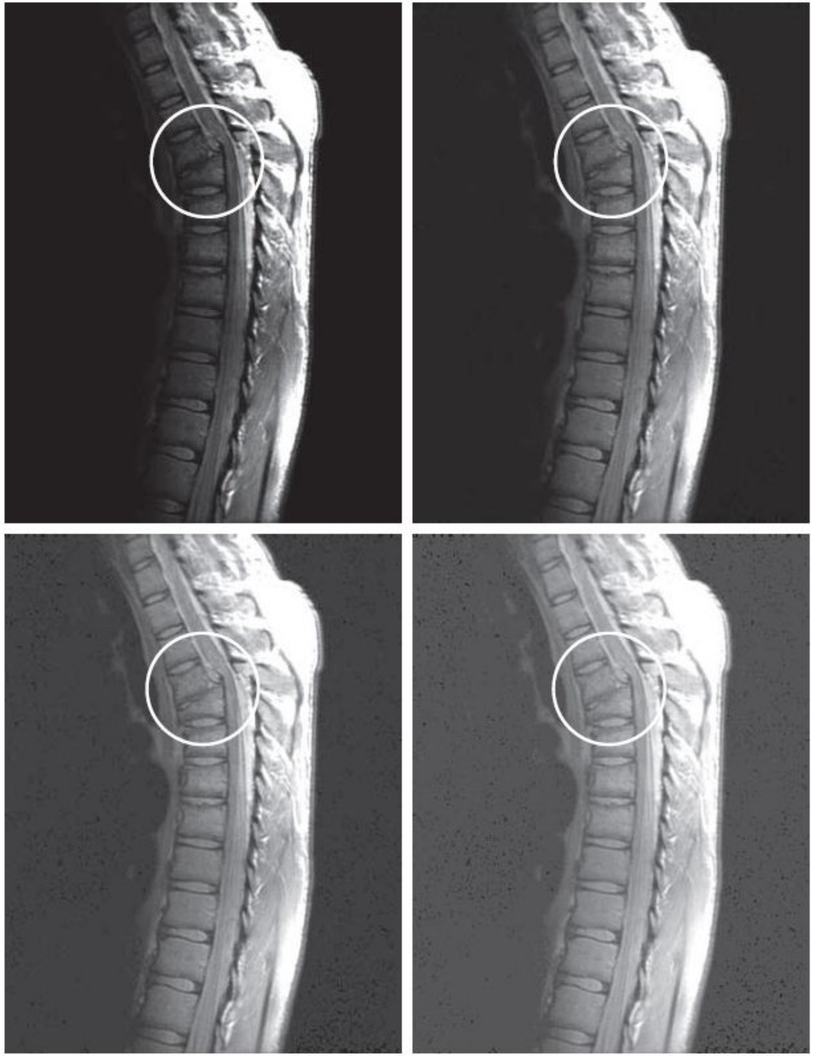


A family of curves gets generated



The response of many devices used for image capture, printing, and display obey a power law. By convention, the exponent in a power-law equation is referred to as gamma

a b
c d



(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle). (b)–(d) Results of applying the transformation in Eq. (3-5) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3, respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

a b
c d

(a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3-5) with $\gamma = 3.0, 4.0$, and 5.0, respectively. ($c = 1$ in all cases.) (Original image courtesy of NASA.)

What should be the value of gamma?

If the Region of Interest (RoI) or Image has more bright values, use gamma>1

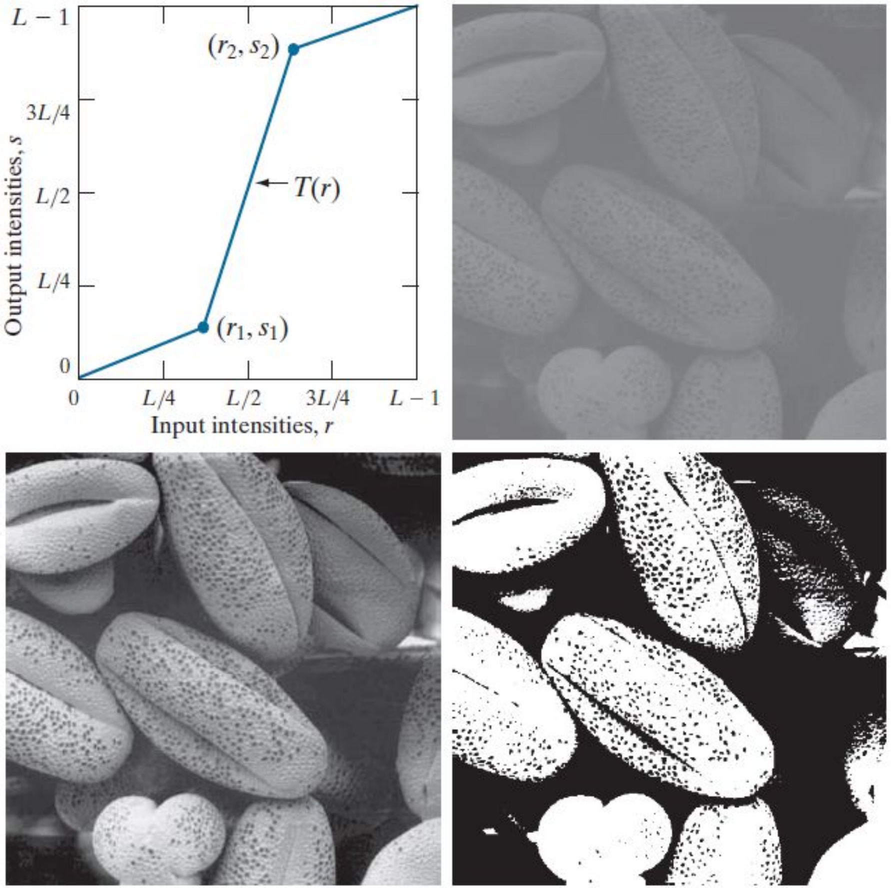
If the Region of Interest (RoI) or Image has more darker values, use gamma<1

Transformation using Complex functions

We can create complex functions, but it requires considerable user input

- Piecewise Linear Transformation
- Intensity-level Slicing
- Bit-plane Slicing

Contrast Stretching



a b
c d

Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

$(r_1, s_1) = (r_{min}, 0)$ and $(r_2, s_2) = (r_{max}, L-1)$

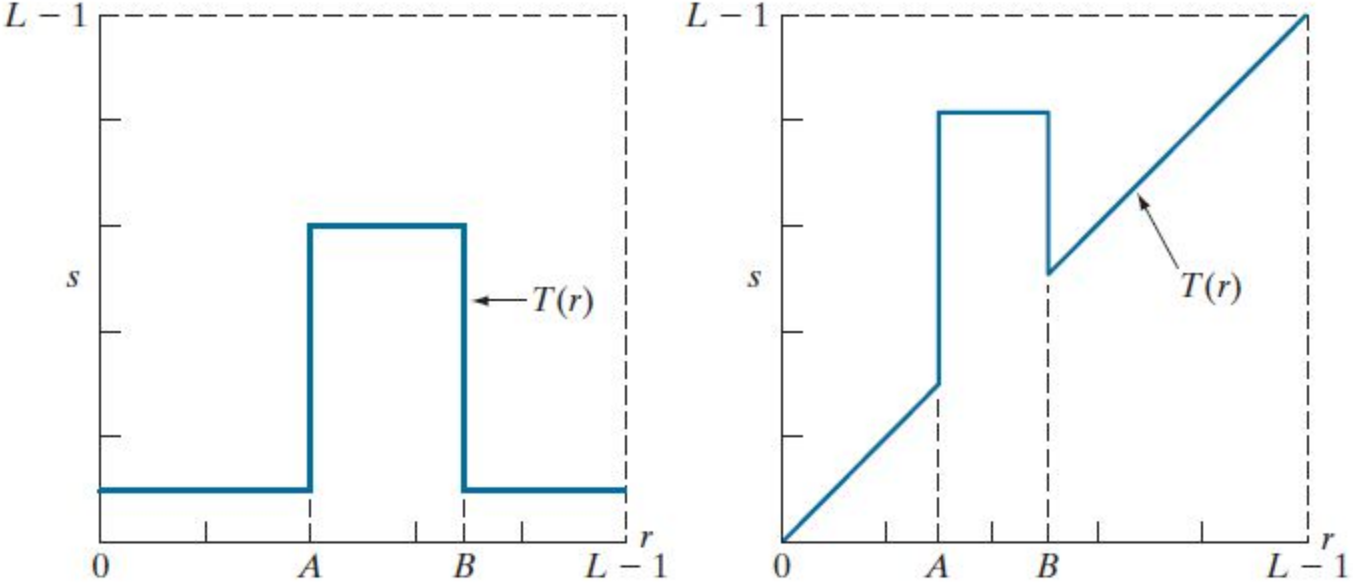
$(r_1, s_1) = (m, 0)$ and $(r_2, s_2) = (m, L-1)$

m=mean pixel value

Intensity Level Slicing: Highlighting or Suppressing certain intensity levels

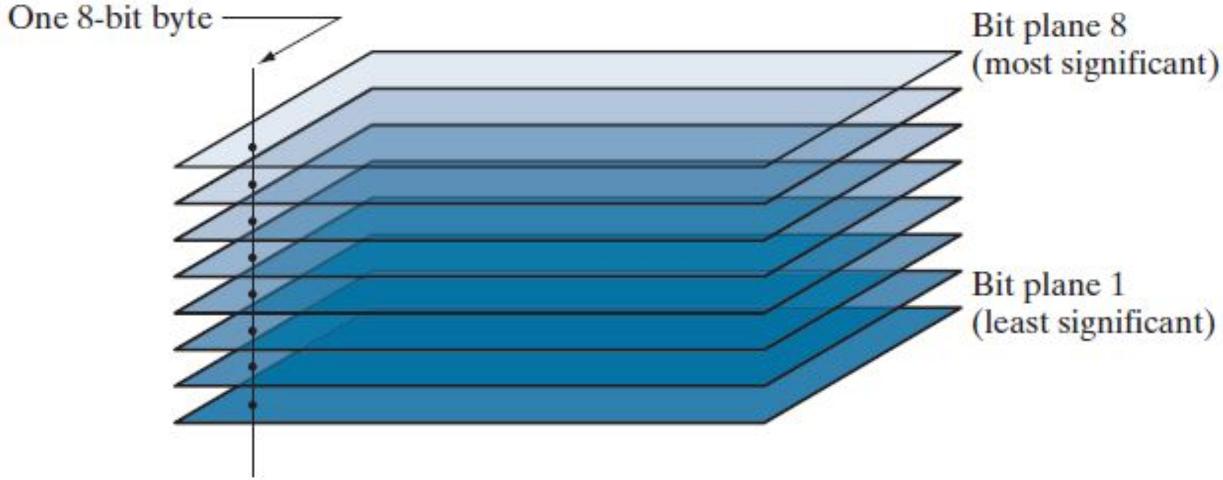
a b

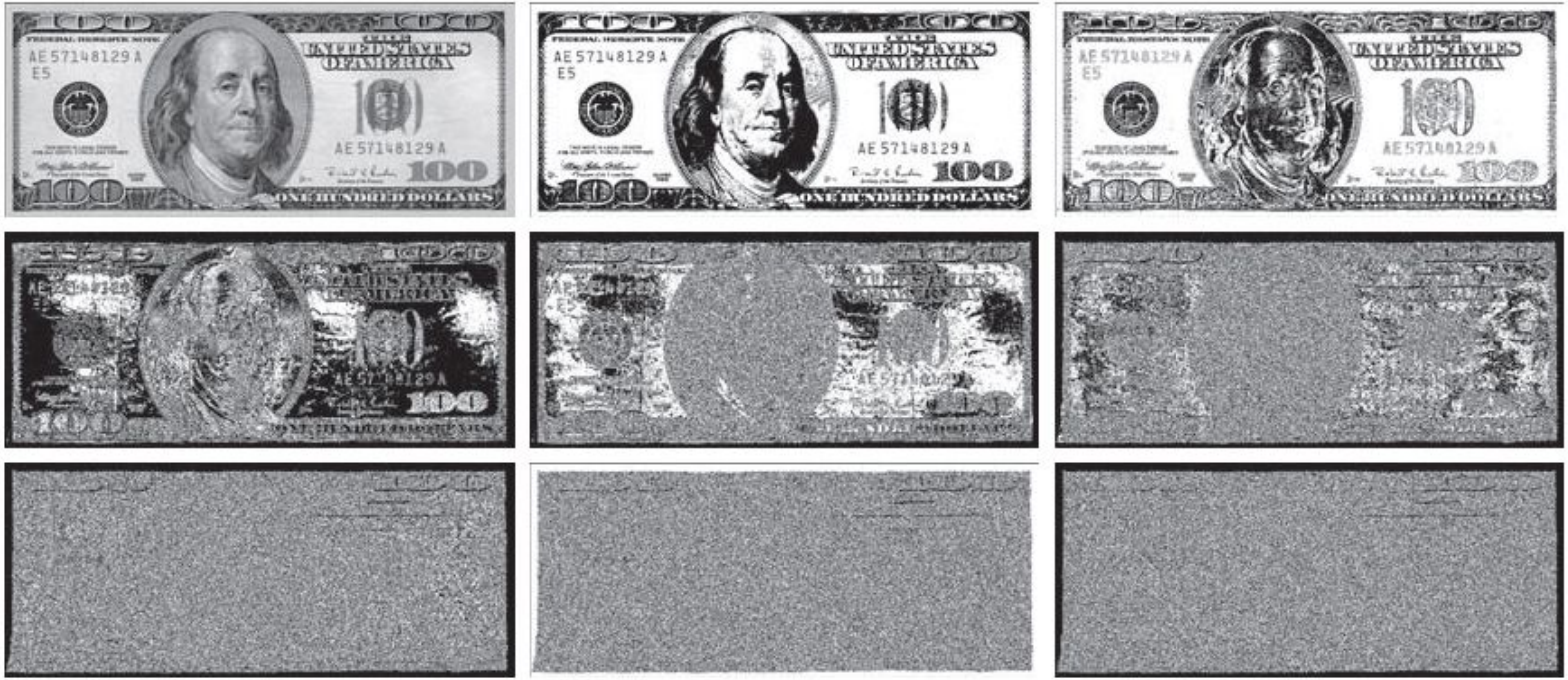
(a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level. (b) This function highlights range $[A, B]$ and leaves other intensities unchanged.



Bit-plane slicing: we can just use highly significant bit planes

Bit-planes of an 8-bit image.





a b c
d e f
g h i

(a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

Reconstructed Image: 128*B8+64*B7+32*B6+16*B5+8*B4+4*B3+2*B2+1*B1

If all 8 planes are available

a b c

Image reconstructed from bit planes: (a) 8 and 7; (b) 8, 7, and 6; (c) 8, 7, 6, and 5.

Bit-plane Slicing

Depending upon different bit planes available, different reconstructed images can be created. If a particular bit plane isn't available, the corresponding term gets removed.